

Part I: LaTeX Debugging Log

Designing Verification Checkpoints to Mitigate Silent Errors in Agentic AI Pipelines

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During the LaTeX conversion process, the file main.tex was compiled using pdflatex and bibtex (three compilation passes) to check whether the document builds correctly and to look for any compilation errors, undefined references, or formatting warnings. The compilation log was checked manually rather than assumed to be error-free. Three problems were found and corrected during this process. Each one is documented below, along with its cause, the suggestion used to fix it, and the final correction applied. No scientific content, equations, or numerical results were changed while fixing these issues.

Table: Errors Found and Fixed

#	Error / Problem	Likely Cause	Prism Suggestion	Final Correction
1	Undefined control sequence: \minimize while compiling the checkpoint-optimization equation.	\minimize is not a real LaTeX or amsmath command. LaTeX has no built-in macro for the word “minimize” as a math operator.	Use \operatorname{} from amsmath to typeset multi-letter operator names in math mode.	Replaced \minimize with \operatorname{minimize} in the checkpoint-selection optimization equation.
2	Package pgfkeys Error: “I do not know the key /tikz/diamond” and “I do not know the key /tikz/aspect”, raised while compiling the risk-adaptive checkpoint flow diagram (Figure 1).	The diamond decision-node shape and its aspect key belong to TikZ’s shapes.geometric library, which was not loaded in the preamble.	Find which TikZ library defines the shape/key being used and load it with \usetikzlibrary{} before it is used.	Added shapes.geometric to the \usetikzlibrary{} line in the preamble. The decision diamonds rendered correctly after recompiling.
3	Overfull \hbox (17.95pt too wide) reported at the lines containing the risk-adaptive checkpoint flow diagram (Figure 1).	The natural width of the TikZ diagram was slightly wider than the text block defined by the page margins.	Scale the figure down to fit \linewidth instead of manually resizing each node.	Wrapped the tikzpicture in \resizebox{\linewidth}{!}{}. The overfull hbox warning no longer appears.

Verification Note

After making all three corrections, the document was recompiled through three pdflatex passes with a bibtex run in between. The final compilation log showed no errors, no undefined citations, and no undefined references. All labelled tables, figures, sections, and equations were checked and confirmed to display the correct numbers in the generated PDF.